LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Practice Pack: Force and Pressure

Chapter / Topic: Topic-based: use with whichever Class 8 book your students follow

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Part A checks the basics. Part B makes you think and apply.
2. Pressure = Force ÷ Area. SI unit: pascal (Pa) = N/m^2.

PART A: BASICS

Q1. What is a force? Give two effects a force can have on an object.

Q2. Name two contact forces and two non-contact forces.

Q3. Why is it easier to cut vegetables with a sharp knife than with a blunt one?

Q4. Why do liquids leak from holes at the bottom of a bottle with more force than from holes near the top?

Q5. What is atmospheric pressure?

PART B: THINK AND APPLY

Q6. A force of 50 N acts on an area of 0.5 m^2. Find the pressure.

Q7. A brick weighs 30 N. Its faces measure 0.2 m × 0.1 m, 0.2 m × 0.05 m and 0.1 m × 0.05 m. On which face should it rest to put the least pressure on the ground? Calculate the pressure in each case.

Q8. Why do school bags have wide straps?

Q9. Why are the walls of a dam made thicker at the bottom?

Q10. A suction hook (rubber sucker) sticks to a smooth wall. Explain why.

Q11. Camels walk easily on sand, but people sink in. Why?

~ Small area → big pressure. Big area → small pressure. Look for it everywhere! ~